Marcos Canedo

Contact Information

Instituto de Matemática Pura e Aplicada (IMPA)
Estrada Dona Castorina 110, Jardim Botânico
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Profesional Website

Education

Ph.D., Mathematics, Pontificia Universidad Católica de Chile, Santiago, Chile — **Sep 2026**

Dissertation Advisor: Prof. Giancarlo Urzúa

Dissertation Title: *On Surfaces with QHD singularities*

M.Sc., Mathematics, Pontificia Universidad Católica de Chile, Santiago, Chile — **Jan 2024**

B.S., Mathematics, Universidad Mayor de San Andrés, La Paz, Bolivia — **June 2021**

Degree conferred with honor.

Employment

Postdoctoral Researcher, IMPA, Rio de Janeiro, Brazil — **Oct 2026–Present**

Research Interests

My research lie in the intersection of birational geometry and deformation theory of algebraic projective surfaces over the complex numbers. I have worked on the construction of $\mathbb{Q}$-Gorenstein smoothings of singular algebraic projective surfaces, with particular interest in degenerations that preserve the Betti numbers and their applications into problems of algebraic geometry.

Honors and Awards

ANID Doctoral Fellowship, — **2022–2026**
Agencia Nacional de Investigación y Desarrollo, Chile

ANID Operational Research Grant, — **2024–2026**
Agencia Nacional de Investigación y Desarrollo, Chile

Graduated with Highest Honors, — **2022**
Universidad Mayor de San Andrés, Bolivia

Teaching Experience

Substitute Instructor, Pontificia Universidad Católica de Chile - Chile

MAT 1610, Cálculo I — **Second Semester 2026**

Summer Course Instructor, Universidad Mayor de San Andrés - Bolivia

Classifying Conics, GEA - Summer School — **February 2026**

Introduction to Affine Algebraic Curves — **February 2025**

Instructor, Pontificia Universidad Católica de Chile - Chile

MAT 1640, Ecuaciones Diferenciales — **First Semester 2025**

MAT 1640, Ecuaciones Diferenciales — **First Semester 2024**

Teaching Assistant, CIMPA, V ELGA - Brazil

Surfaces with Wahl Singularities and their Many Applications — **12–23 Aug 2024**

Teaching Assistant, Universidad mayor de San Andrés - Bolivia

MAT 111, Álgebra I — **Second Semester 2021**

MAT 111, Álgebra I — **First Semester 2021**

MAT 111, Álgebra I	**Second Semester 2020**
MAT 111, Álgebra I	**First Semester 2020**
MAT 123, Geometry II	**Second Semester 2019**
MAT 113, Geometry I	**Second Semester 2019**
MAT 123, Geometry II	**First Semester 2019**
MAT 113, Geometry I	**First Semester 2019**
MAT 113, Geometry I	**Second Semester 2018**

Professional Activities, Outreach, and Service

Summer Course Coorganizer, Universidad Mayor de San Andrés - Bolivia

Geometría en las Alturas, **February 2026**
Intensive international 3 day school of algebraic geometry ∼60 participants

Introduction to affine algebraic curves, **February 2025**
Intensive 2 week course of algebraic geometry ∼30 participants

Seminar Coorganizer, Pontificia Universidad Católica de Chile - Chile

Seminario de Estudiantes de Postgrado en Matemáticas, **2023–2024**

Computer Skills

Experience in Mathematica, Python, Julia. Markup languages: LaTeX, HTML, Markdown.

Software—My contributions are available on my GitHub profile.

- Calculation of $\mathbb{Q}$HD discrepancies.
- Computation of complete M-resolutions of cyclic quotient singularities.

Publications in Progress

4. **Canedo, M.**, Urzúa, G. (2026) *On $\mathbb{Q}$HD Horikawa Surfaces.*
3. **Canedo, M.**, (2026) *M-modifications of two-dimensional log-canonical singularities.*

Submitted Publications

4. **Canedo, M.**, Urzúa, G., (2026) *Degenerations of the complex projective plane with only rational singularities*, [arXiv:2607.19348].
3. **Canedo, M.**, Urzúa, G., (2026) *Rational homology disk degenerations of elliptic surfaces*, [arXiv:2605.06668].

Poster Presentations

2. Iberoamerican Congress on Singularities — 8 Dec 2025
Small Surfaces with WQHD singularities — Valparaíso, Chile

1. Math-Amsud School on Geometry — 4-7 Dec 2023
W-Surfaces and their MMP — El Quisco, Chile

Contributed Talks

3. Conference: Algebraic Varieties and their Singularities, Universidad de Talca — 30 Nov 2025
Small surfaces with only WQHD singularities — Talca, Chile

2. Sesión de Geometría Algebraica, Sociedad Matemática de Chile — 25 Nov 2025
Small surfaces with only WQHD singularities — Vicuña, Chile

1. Chungnam National University — 30 Oct 2025
Classification of Small surfaces with only WQHD singularities — Daejeon, South Korea

Invited Talks

8. Universidad de la Frontera — 18 Nov 2025
   Small surfaces with only WQHD singularities — Temuco, Chile
7. Institute for Basic Science — 13 Nov 2025
   Small surfaces with only WQHD singularities — Daejeon, South Korea
6. Seminario de Estudiantes de Postgrado - PUC — 16 Oct 2025
   ¿Qué es una Small Surface? — Santiago, Chile
5. Seminario de Estudiantes de Postgrado en Matemáticas - PUC — 30 Oct 2024
   Horikawa Surfaces — Santiago, Chile
4. Seminario del Grupo de Geometría Ecuador — 28 Oct 2024
   Horikawa Surfaces — ONLINE
3. Seminario de Estudiantes de Postgrado - UT — 4 Oct 2024
   Horikawa Surfaces — Talca, Chile
2. Taller: "Experimenta la Topología a través de la Arcilla" - PUC — 25-26 Sep 2024
   Development, Introduction to the Workshop — Santiago, Chile
1. Club de Matemática UC — 31 Mar 2023
   Música y Matemáticas: Acciones Musicales del Grupo Dihedral — Santiago, Chile

Seminar Talks

Seminar of Algebraic Geometry, Pontificia Universidad Católica de Chile

15. Classification of Small Surfaces with only WQHD singularities — 9 Oct 2025
14. Rational Homology Disk Smoothing of $W_{P,q,r}$ part 2 — 6 Jun 2025
13. Rational Homology Disk Smoothing of $W_{P,q,r}$ part 1 — 29 May 2025
12. Deformaciones de superficies algebraicas y obstrucciones — 3 April 2025
11. Superficies singulares y no singulares — 27 March 2025
10. Fracciones continuas negativas, cadenas de Wahl, fracciones del 0 — 20 March 2025
9. La Conjetura QHD y su intento de demostración — 9 Apr 2024
8. Singularidades Terminales — 31 May 2023
7. Teorema del Cono: Singular VS No Singular-III — 24 May 2023
6. Teorema del Cono: Singular VS No Singular-II — 17 May 2023
5. Teorema del Cono: Singular VS No Singular-I — 10 May 2023
4. Geometría Birracional de Superficies — 3 May 2023
3. Geometría Birracional de Curvas y Superficies-II — 12 Apr 2023
2. Geometría Birracional de Curvas y Superficies-I — 5 Apr 2023
1. Introducción a la Geometría Birracional — 29 Mar 2023

References

Giancarlo A. Urzúa, Full Professor, Pontificia Universidad Católica de Chile
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